North South University

Department of Electrical and Computer Engineering

Project title: SkillNest- Online Learning Platform

Submitted To: AKM Iqtidar Newaz (IqN)

Group ID:

Group Members:

1. **Urbana Andalib 2322005042**
2. **Sayeda Ruhia Jannat Ibra 2323698042**
3. **Nuha Nur Sikder 2321115042**
4. **Syed Ashfaq Shafin 2312845042**

Section: 07

Project Overview:

SkillNest is an online Learning Management System designed to support the delivery and management of educational content. The platform provides teachers and students with the tools they need to create, access, and track learning material in one place.

Teachers can create and manage courses, lessons, and assessments. Students can enroll in courses, work through content at their own pace, and monitor their progress over time. The platform supports multimedia content and interactive quizzes to keep learning structured and engaging.

Access control is handled through a role-based system, giving teachers, students, and administrators appropriate permissions within the platform. Built-in notifications keep users informed of relevant activity, such as new course content or quiz results.

SkillNest also includes a basic analytics feature that tracks student performance, quiz completion rates, and learning trends. Teachers and administrators can use this data to review course effectiveness and identify where students may need additional support.

End Users:

Young Learners

Goal: Learn online in an engaging manner.

Role: Actively engage with the platform's learning modules, challenges, and quizzes.

Educators

Goal: Enhance their teaching methods and track student progress.

Role: Supplement their teaching with the platform and possibly set assignments or challenges.

Admins

Goal: Ensure the smooth running of the platform.

Role: Handle user complaints, manage accounts, and oversee general platform health

Functional Requirements:

User Authentication and Role Management: The system shall allow users to register and log in with role-based access, granting teachers, students, and admins appropriate permissions within the platform.

Course and Lesson Management: The system shall allow teachers to create, update, and delete courses and lessons, including support for multimedia content.

Student Enrollment and Progress Tracking: The system shall allow students to enroll in courses, work through content at their own pace, and view their progress over time.

Quiz Creation and Assessment: The system shall allow teachers to create interactive quizzes and students to attempt them, with results recorded and accessible to both parties.

Admin Dashboard and Platform Oversight: The system shall provide admins with tools to manage user accounts, handle complaints, and monitor overall platform health.

Non-Functional Requirements:

Performance: Ensures the system remains responsive, even with many users, to provide a seamless learning experience.

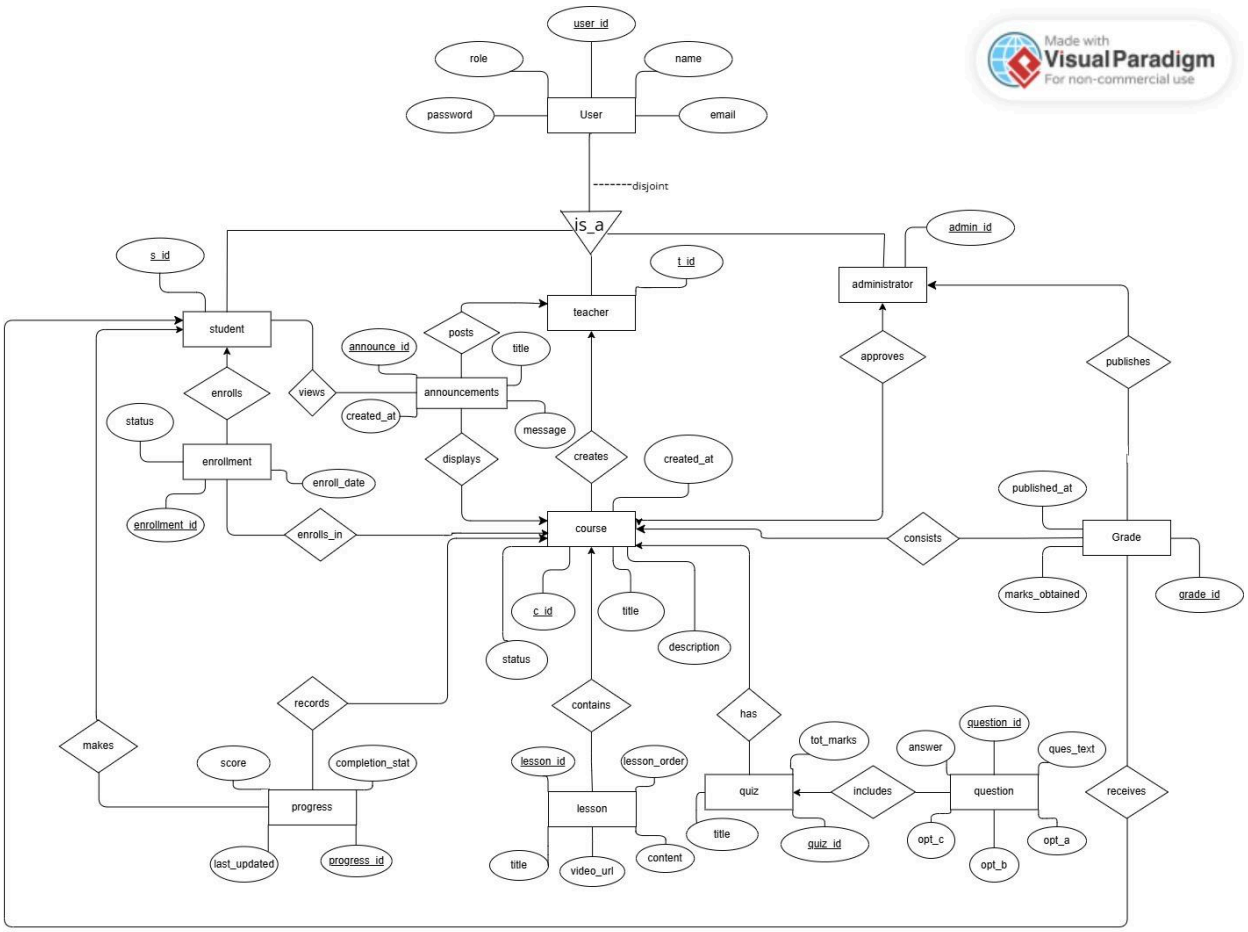
Usability: Facilitates learning by making the platform intuitive and easy to navigate.

Security: Protects user data, especially crucial given the potential presence of minors.

Ethical and Inclusive: Guarantees that the platform remains accessible, fair, and equitable for all users.

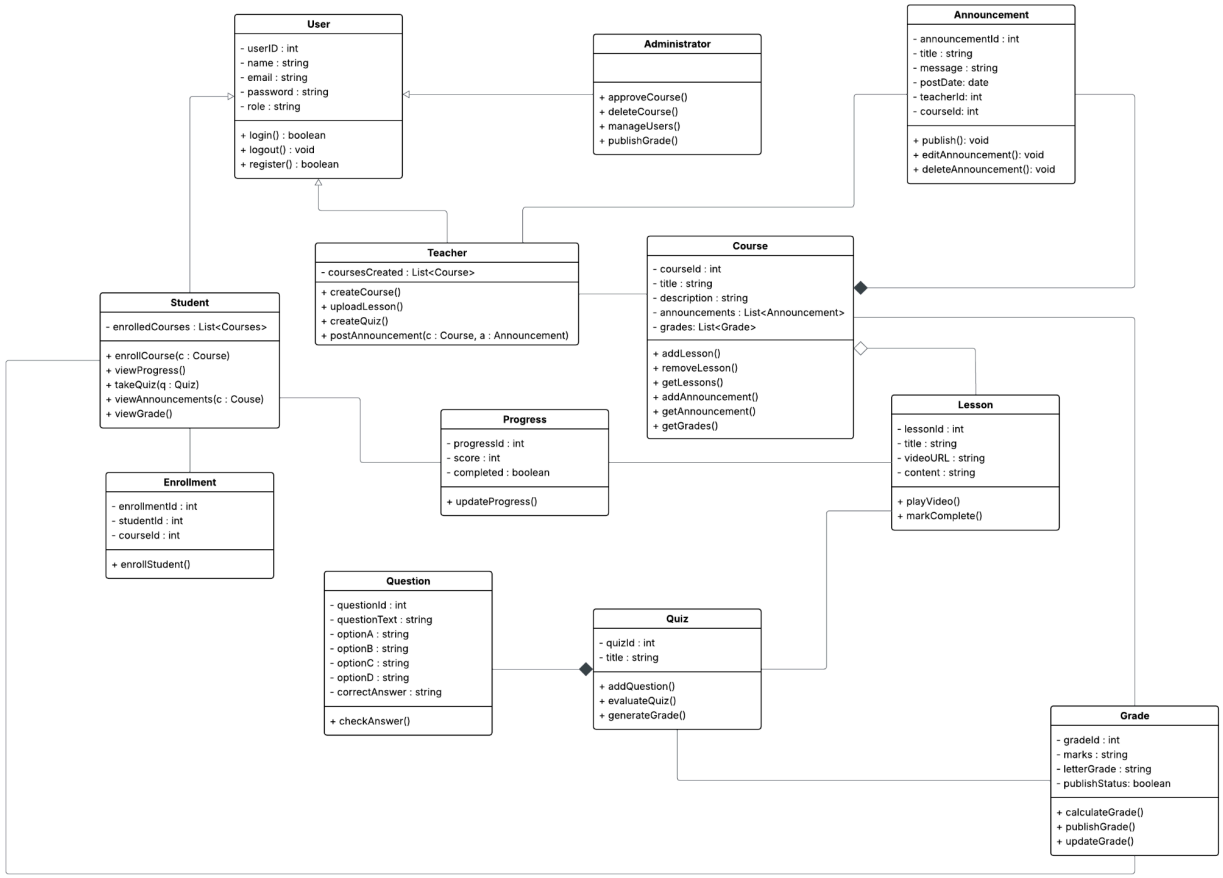
Reliability: Ensures the platform is dependable, fostering user trust

1. Entity Relationship Diagram:

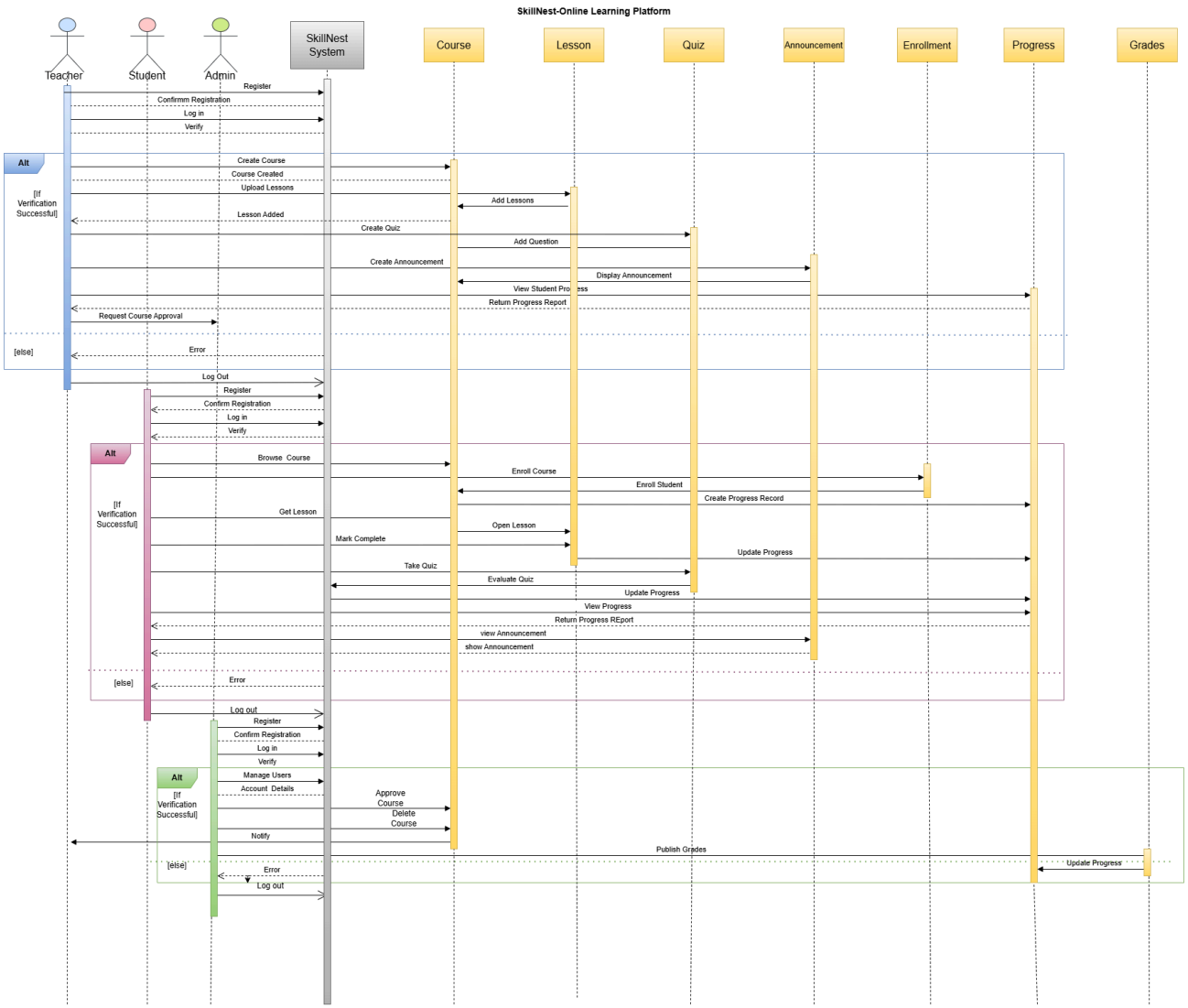


2. Use-case diagram:

3. Class Diagram:



4. Sequence diagram:



GitHub Repository:

[ibraBzz/SkillNest-Online-Learning-Platform: CSE327 Software Engineering Project]